

Qi Xu

Multimodal Large Language Models ■ Video Understanding ■ Multimodal Post-Training

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EDUCATION



Westlake University

PhD Student in Artificial Intelligence
Co-supervised by Peidong Liu and Chi Zhang

2026 – Present



Wuhan University

Master in Pattern Recognition and Intelligent Systems
Research: Multimodal Learning, Computer Vision, 3D Reconstruction

2023 – 2026



Wuhan University

B.S. in Spatial Information and Digital Technology

2019 – 2023

RESEARCH PROFILE

My research focuses on **multimodal large language models for video understanding and spatio-temporal reasoning**. I work on multimodal data construction, SFT/RL post-training, and evidence-grounded evaluation, with applications in temporal grounding and embodied captioning. My broader interests include connecting vision, language, and 3D scene representations to improve spatial understanding and reasoning about the physical world.

EXPERIENCE



ByteDance, Data-AML-Doubao

Research Intern; Mentored by Ke Wang and Xiongcai Luo

Aug 2026 – Present

- Designed and scaled **synthetic data pipelines** for the **embodied captioning** and **Doubao automotive** projects.
- Contributed to **Doubao-Seed-Embodied-Caption** model training.



ByteDance, TikTok Global Live

Research Intern, LLM Strategy; Mentored by Xiangtai Li

Sep 2025 – Mar 2026

- Co-developed **video MLLMs for temporal grounding**, focusing on **SFT/RL post-training**, reward design, and evaluation.
- Built **multimodal training data and annotation pipelines** for temporally grounded video understanding.

SELECTED PUBLICATIONS

First Author: first-listed author. * Equal contribution. † Corresponding author.



SIU3R: Simultaneous Scene Understanding and 3D Reconstruction Beyond Feature Alignment

NeurIPS 2025 **First Author** **Spotlight** 160+ GitHub Stars 10+ Citations

[Project Page](#) [ArXiv](#) [Code](#)

2025

Qi Xu*, Dongxu Wei*, Lingzhe Zhao, Wenpu Li, Zhangchi Huang, Shunping Ji, Peidong Liu

Introduced a shared pixel-aligned 3D representation for simultaneous reconstruction and scene understanding from unposed images, providing a unified spatial state for geometry and semantics.



Towards One-to-Many Temporal Grounding

ICML 2026 **First Author** [Project Page](#) [ArXiv](#)

2026

Qi Xu*, Yue Tan*, Shihao Chen, Jiahao Meng, Anran Wang, Shunping Ji, Hao Fei, Xiangtai Li

Introduced the first systematic solution for OMTG, featuring a 56k-sample dataset and RL training with CoT-based rewards; outperformed Gemini 2.5 Pro by 15.85%.



VideoZeroBench: Probing the Limits of Video MLLMs with Spatio-Temporal Evidence Verification

arXiv 2026 [Project Page](#) [ArXiv](#) [Code](#)

Jiahao Meng, Tan Yue, **Qi Xu**, Haochen Wang, Zhongwei Ren, Weisong Liu, Yuhao Wang, Renrui Zhang, Yunhai Tong, Haodong Duan

A hierarchical long-video QA benchmark that verifies model answers with temporal intervals and spatial boxes under a five-level evidence protocol.

SELECTED PUBLICATIONS (CONTINUED)



Watch, Remember, Reason: Human-View Video Understanding with MLLMs

arXiv 2026 [ArXiv](#) [GitHub](#)

Jiahao Meng, Yue Tan, **Qi Xu**, Kuan Gao, Weisong Liu, Yanwei Li, Xiangtai Li, Lingdong Kong, Haochen Wang, 2026
Qianyu Zhou, Jiangning Zhang, Guangliang Cheng, Yunhai Tong, Lu Qi, Mingsuan Yang
Surveyed human-view video understanding with MLLMs through watching, remembering, and reasoning abilities for long, multimodal, and knowledge-intensive video scenarios.



HiCI: Hierarchical Construction-Integration for Long-Context Attention

ICML 2026 [ArXiv](#) [Code](#)

Xiangyu Zeng, **Qi Xu**, Yunke Wang, Chang Xu 2026
Efficient long-context modeling using hierarchical attention, extending LLaMA-2 to 100K tokens with only 5.5% additional parameters.



Any 3D Scene is Worth 1K Tokens: 3D-Grounded Representation for Scene Generation at Scale

arXiv 2026 [Co-First Author](#) [Project Page](#) [ArXiv](#) [Code](#)

Dongxu Wei*, **Qi Xu***, Zhiqi Li, Hangning Zhou, Cong Qiu, Hailong Qin, Mu Yang, Zhaopeng Cui, Peidong Liu 2026
Introduced a compact 1K-token implicit 3D latent representation and 3D Diffusion Transformer for scalable scene generation.



SupIR-GS: Thermal Infrared Super-Resolution Novel View Synthesis with Imaging-Calibrated 3D Gaussian Splatting

ECCV 2026 [Corresponding Author](#) [ECCV](#)

Jin Liu, Haodong Li, Jiagang Chen, Dabin leng, Jiguang Li, Zhao Huang, Xiaoshuai Zhang, **Qi Xu**[†], Zhiwen Zheng, 2026
Xingru Huang[†]
Reconstructs high-resolution 3D thermal scenes from low-resolution inputs via physics-informed degradation modeling.

Mentorship: Provided the core idea, guided method design and experiment planning, and led paper writing and revision.



Physically Grounded Dual-Opacity Gaussian Splatting for Joint RGB-TIR Reconstruction

ECCV 2026 [Corresponding Author](#) [ECCV](#)

Jin Liu, Dabin leng, Jiagang Chen, Haodong Li, Jiguang Li, Zhao Huang, Xiaoshuai Zhang, Zhiwen Zheng, Xingru Huang[†], **Qi Xu**[†] 2026
A unified RGB-TIR reconstruction framework resolving cross-spectral visibility conflicts through thermal field modeling.

Mentorship: Proposed the core idea, coordinated experiments and ablations, and led paper writing and revision.



E-MoFlow: Learning Egomotion and Optical Flow from Event Data via Implicit Regularization

NeurIPS 2025 [Project Page](#) [ArXiv](#) [Code](#)

Wenpu Li, Bangyan Liao, Yi Zhou, **Qi Xu**, Pian Wan, Peidong Liu 2025
An unsupervised framework for joint ego-motion and optical flow estimation using implicit neural representations and differential geometric constraints.

SKILLS

Research: Multimodal SFT/RL post-training, video understanding, spatio-temporal grounding, synthetic data construction, and evidence-grounded evaluation.

Direction: Long-context multimodal reasoning, spatially grounded MLLMs, and embodied scene understanding.

Engineering: Python, PyTorch, distributed training/evaluation, synthetic data and annotation pipelines, and large-scale experimental analysis; familiar with C++ and Linux development.

Writing & Leadership: Leading research projects from problem formulation to implementation, benchmarking, analysis, paper writing, and public release.